

MODULE 09: LC CIRCUIT & MAXWELL EQUATIONS

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LECTURE 21

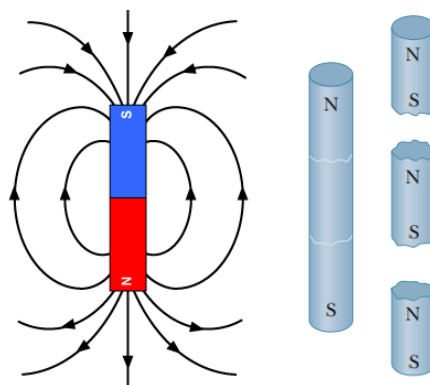
OUTLINE:

- Gauss' Law for Magnetic Fields
- Induced Magnetic Fields
- Ampere-Maxwell Law
- Displacement Current

GAUSS' LAW FOR MAGNETIC FIELDS

Magnetic pole refers to region at each end of a magnet where the external magnetic field is strongest.

- One end of the magnet is a *source* of the field (the field lines diverge from it) and the other end is a *sink* of the field (the field lines converge toward it).
- By convention, the source is called the *north pole* of the magnet and the sink the *south pole*.
- The magnet, with its two poles, is an example of a magnetic dipole.
- Even if a bar magnet is broken into pieces, we end up having fragment with a north pole and a south pole.
- The simplest magnetic structure that can exist is a magnetic dipole.
- ***Magnetic monopoles do not exist (as far as we know).***



Gauss' Law for Magnetic Fields, also known as Gauss' Law for Magnetism, is a fundamental principle in electromagnetism that relates the magnetic field to the distribution of magnetic charge (magnetic monopoles) or the flow of electric current.

Gauss' Law for Magnetic Fields: The law states that the net magnetic flux Φ_B through any closed Gaussian surface is zero. Gauss' law for magnetic fields is a formal way of saying that magnetic monopoles do not exist. Mathematically, this can be expressed as:

$$\Phi_B = \oint \vec{B} \cdot d\vec{A} = 0$$

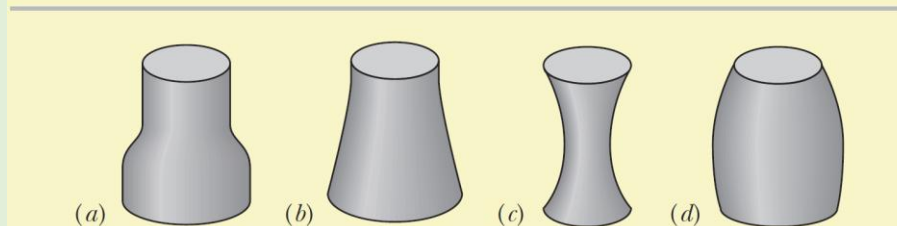
where integration sign represents the surface integral over a closed surface, $\vec{B}$ is the magnetic field vector, and $d\vec{A}$ is an infinitesimal vector area element.

Gauss' law for magnetic fields says that there is no net magnetic flux through the surface because no net "magnetic charge" (individual magnetic poles) can be enclosed by the surface.

- The simplest magnetic structure that can exist and thus be enclosed by a Gaussian surface is a magnetic dipole, which consists of both a source and a sink for the field lines.
- Thus, there must always be equal number of magnetic fluxes into the surface and out of it, and thus, the net magnetic flux must always be zero.

CHECKPOINT: The figure here shows four closed surfaces with flat top and bottom faces and curved sides. The table gives the areas A of the faces and the magnitudes B of the uniform and perpendicular magnetic fields through those faces; the units of A and B are arbitrary but consistent. Rank the surfaces according to the magnitudes of the magnetic flux through their curved sides, greatest first.

Surface	A_{top}	B_{top}	A_{bot}	B_{bot}
a	2	6, outward	4	3, inward
b	2	1, inward	4	2, inward
c	2	6, inward	2	8, outward
d	2	3, outward	3	2, outward



ANSWER: $d > b > c > a$ (zero)

magnetic flux through their curved surface (a): $|\Phi_B| = |(6)(2) + (-3)(4)| = |0| = 0$

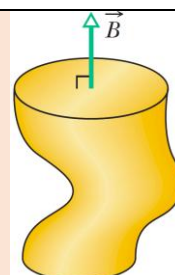
magnetic flux through their curved surface (b): $|\Phi_B| = |(-1)(2) + (-2)(4)| = |-10| = 10$

magnetic flux through their curved surface (c): $|\Phi_B| = |(6)(2) + (-8)(2)| = |-4| = 4$

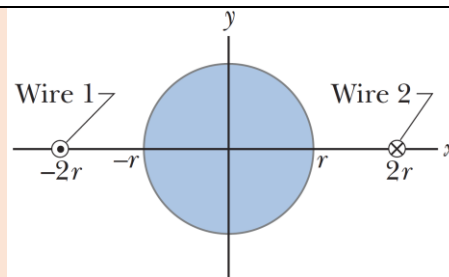
magnetic flux through their curved surface (d): $|\Phi_B| = |(3)(2) + (2)(3)| = |12| = 12$

PROBLEM 32-01: The magnetic flux through each of five faces of a die (singular of “dice”) is given by $\Phi_B = \pm NWb$, where N ($= 1$ to 5) is the number of spots on the face. The flux is positive (outward) for N even and negative (inward) for N odd. What is the flux through the sixth face of the die?

PROBLEM 32-02: Figure 32-27 shows a closed surface. Along the flat top face, which has a radius of 2.0 cm, a perpendicular magnetic field of magnitude 0.30 T is directed outward. Along the flat bottom face, a magnetic flux of 0.70 mWb is directed outward. What are the (a) magnitude and (b) direction (inward or outward) of the magnetic flux through the curved part of the surface?



PROBLEM 32-04: Two wires, parallel to a z axis and a distance $4r$ apart, carry equal currents i in opposite directions, as shown in figure. A circular cylinder of radius r and length L has its axis on the z axis, midway between the wires. Use Gauss' law for magnetism to derive an expression for the net outward magnetic flux through the half of the cylindrical surface above the x axis.



INDUCED MAGNETIC FIELDS

Induced magnetic fields refer to the creation of a magnetic fields in a materials or space when it is subjected to a changing electric field. Induced magnetic fields arise due to **time-varying electric fields** or **moving charges** (currents).

The equation governing the induction of a magnetic field, is known as **Maxwell's law of induction** after James Clerk Maxwell, and can be written as,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

Here $\vec{B}$ is the magnetic field induced along a closed loop by the changing electric flux Φ_E in the region encircled by that loop.

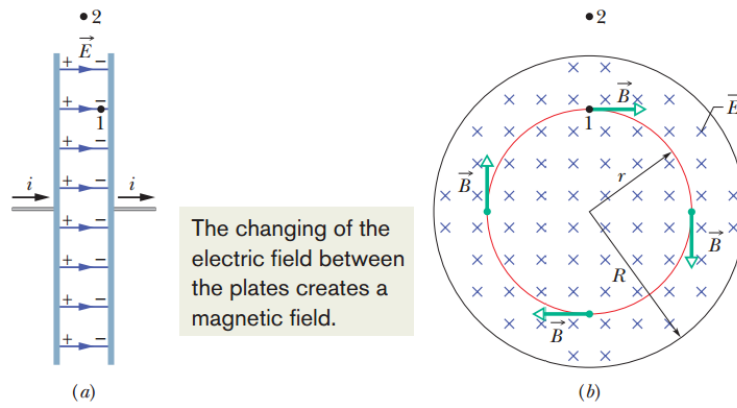
Charging a Capacitor:

Let us consider a charging of a parallel-plate capacitor having circular plates of area A , where the charges on the capacitor (Fig. *a*) are being increased at a steady rate by a constant current i in the connecting wires.

- Resulting the magnitude of the electric field ($\vec{E}$) between the plates increasing at a steady rate.

The front view of the capacitor plates (Fig. *a*) and view from between the plates is presented in Fig. *b*. The electric field is directed into the page as showed in Fig. *b*. Let us consider a circular loop through **point 1** in Fig. *a* and Fig. *b*, a loop that is concentric with the capacitor plates and has a radius smaller than that of the plates.

- As the electric field through the loop is changing, so thus the electric flux through the loop must also be changing.
- This changing electric flux induces a magnetic field ($\vec{B}$) around the loop.
- Direction of this induced magnetic field follows right hand curl law.
- This magnetic field has same magnitude at every point around the loop and has circular symmetry about the *central axis* of the capacitor plates (axis extending from one plate center to other).



Now if a larger loop through **point 2** outside the plates is considered in *Fig. a* and *Fig. b*. It has been observed that a magnetic field is induced around that loop as well. Thus, while the electric field is changing, magnetic fields are induced between the plates, both inside and outside the gap. When the electric field stops changing, these induced magnetic fields disappear.

AMPERE–MAXWELL LAW

The Ampere-Maxwell law, also known as Ampere's circuital law with Maxwell's addition, relates the magnetic field around a closed loop to the electric current passing through that loop and the rate of change of the electric flux through the surface enclosed by the loop.

From Ampere's law we know that the magnetic field around a closed loop is directly proportional to the electric current passing through any surface bounded by that loop. Therefore,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc}$$

Here i_{enc} is the current encircled by the closed loop.

Again, from Maxwell's law of induction we know,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt}$$

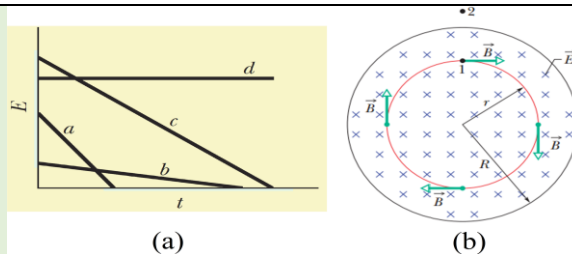
Here $\vec{B}$ is the magnetic field induced along a closed loop by the changing electric flux Φ_E in the region encircled by that loop.

Thus, the Ampere-Maxwell law can be stated by combining the two equations as follows,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{enc} + \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} = \mu_0 \iint \vec{j} \cdot d\vec{A} + \mu_0 \epsilon_0 \frac{d}{dt} \left(\iint \vec{E} \cdot d\vec{A} \right)$$

where $\oint \vec{B} \cdot d\vec{s}$ represents the line integral of the magnetic field $\vec{B}$ around a closed loop, $d\vec{s}$ is an infinitesimal vector element along the loop, ϵ_0 is the vacuum permittivity, μ_0 is the vacuum permeability, $\vec{j}$ is the electric current density, $\vec{E}$ is the electric field, and the second term involves the time derivative of the electric flux ($\oint \vec{E} \cdot d\vec{s}$) passing through the surface.

CHECKPOINT: The figure shows graphs of the electric field magnitude E versus time t for four uniform electric fields, all contained within identical circular regions as in Fig. b . Rank the fields according to the magnitudes of the magnetic fields they induce at the edge of the region, greatest first.



ANSWER: a, c, b, d (zero)

PROBLEM: A parallel-plate capacitor with circular plates of radius R is being charged as in Fig.

(a) Derive an expression for the magnetic field at radius r for the case $r \leq R$.

(b) Evaluate the field magnitude B for $r = \frac{R}{5} = 11.0 \text{ mm}$ and $\frac{dE}{dt} = 1.50 \times 10^{12} \text{ V/m} \cdot \text{s}$.

(c) Derive an expression for the induced magnetic field for the case $r \geq R$.

ANSWER: (a) A magnetic field can be set up by a current and by induction due to a changing electric flux; both effects are included. According to the Maxwell's equation we know,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} \dots \dots \dots (1)$$

The magnetic field $\vec{B}$ at all points along the loop with radius $r \leq R$ is tangent to the loop, as is the path element $d\vec{s}$. Thus, $\vec{B}$ and $d\vec{s}$ are either parallel or antiparallel at each point of the loop. For simplicity, assume they are parallel. Therefore, we get,

$$\oint \vec{B} \cdot d\vec{s} = \oint B ds \cos 0^\circ = \oint B ds$$

Due to the circular symmetry of the plates, we can also assume that $\vec{B}$ has the same magnitude at every point around the loop. Thus, B can be taken outside the integral on the right side of the above equation. The integral that remains is $\oint ds$, which simply gives the circumference $2\pi r$ of the loop. The left side of equation (1) gives $(B)(2\pi r)$.

We assume that the electric field $\vec{E}$ is uniform between the capacitor plates and directed perpendicular to the plates. Then the electric flux Φ_E through the Amperian loop is EA , where A is the area encircled by the loop within the electric field. Thus, the right side of Eq. 1 is $\mu_0 \epsilon_0 \frac{d(EA)}{dt}$.

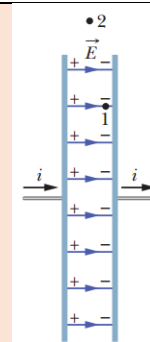
Substituting these values in equation (1) we get,

$$(B)(2\pi r) = \mu_0 \epsilon_0 \frac{d(EA)}{dt}$$

$$(B)(2\pi r) = \mu_0 \epsilon_0 A \frac{dE}{dt} \dots \dots \dots (2) \quad [\because A \text{ is a constant}]$$

The area A that is encircled by the Amperian loop within the electric field is the *full* area πr^2 of the loop because the loop's radius r is less than (or equal to) the plate radius R . Substituting πr^2 for A in Eq. (2) leads to, for $r \leq R$,

$$B = \frac{\mu_0 \epsilon_0 r}{2} \frac{dE}{dt}$$



(b) We have, $B = \frac{1}{2} \mu_0 \epsilon_0 r \frac{dE}{dt}$

$$B = \frac{1}{2} (4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}) (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2) (11.0 \times 10^{-3} \text{ m}) (1.50 \times 10^2 \text{ V/m} \cdot \text{s})$$

$$B = 9.18 \times 10^{-8} \text{ T}$$

(c) Here, we use an Amperian loop with a radius r that is greater than the plate radius R , to evaluate B outside the capacitor. Evaluating the left and right sides of Eq. (1) leads to Eq. (2). The electric field exists only between the plates, not outside the plates. Thus, the area A that is encircled by the Amperian loop in the electric field is *not* the full area πr^2 of the loop. Rather, A is only the plate area πR^2 .

Substituting πR^2 for A in Eq. (2) and solving the result for B give us, for $r \geq R$,

$$B = \frac{\mu_0 \epsilon_0 R^2}{2r} \frac{dE}{dt}$$

This equation tells us that, outside the capacitor, B decreases with increased radial distance r , from a maximum value at the plate edges (where $r = R$).

DISPLACEMENT CURRENT

A displacement current refers to the rate of change of the electric displacement field with time. It represents an additional current that arises from a changing electric field. Unlike conduction current, displacement current does not involve the actual motion of electric charges.

- It is a vector quantity.
- It is measured in Ampere (A).
- It is not produced by the flow of electrons, but by the change in the electric field.
- It does not follow Ohm's law.
- Displacement current (i_d) is a time-varying electric displacement field, symbolized by the term $\epsilon_0 \left(\frac{dE}{dt} \right)$ in Ampere-Maxwell's law, where ϵ_0 is the vacuum permittivity and $\frac{dE}{dt}$ represents the rate of change of the electric field with respect to time.

From Ampere-Maxwell's law we know,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} + \mu_0 i_{enc}$$

By comparing the two terms on the right side of equation, the product $\epsilon_0 \frac{d\Phi_E}{dt}$ has been treated as being a fictitious current called the **displacement current** i_d . Therefore,

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt}$$

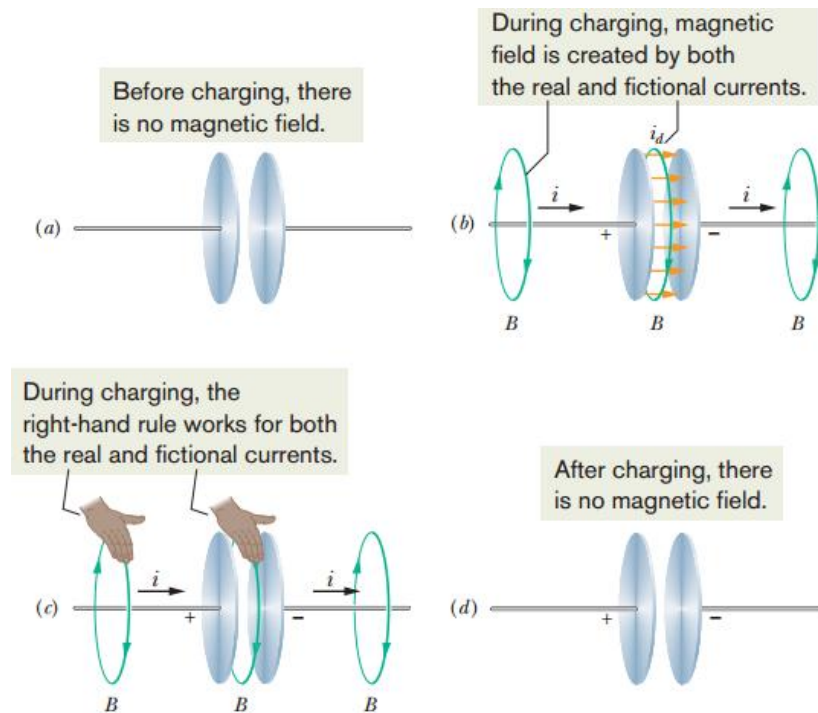
Thus, we can write,

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 i_{d,enc} + \mu_0 i_{enc}$$

in which $i_{d,enc}$ is the displacement current that is encircled by the integration loop.

Example:

Let us consider a charging capacitor with circular plates, as presented in Fig. a. The current (i) is charging the plates, leading to the changes in the electric field between the plates. The fictitious displacement current i_d forms between the plates due to changing field $\vec{E}$ between the plates.



The charge q on the plates at any time is related to the magnitude E of the field between the plates at that time and the plate area A by,

$$q = CV = \frac{\epsilon_0 A}{d} V = \epsilon_0 A E \dots \dots \dots (1) \quad \left[\text{since, } C = \frac{\epsilon_0 A}{d}, C = \frac{q}{V} \text{ \& } E = \frac{V}{d} \right]$$

The real current i , can be obtained by differentiating equation with respect to time,

$$i = \frac{dq}{dt} = \epsilon_0 A \frac{dE}{dt} \dots \dots \dots (2)$$

We know the displacement current (i_d) is given by,

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt} \dots \dots \dots (3)$$

Assuming that the electric field between the two plates is uniform, and the electric flux $\Phi_E = EA$. Then, we get,

$$i_d = \epsilon_0 \frac{d\Phi_E}{dt} = \epsilon_0 \frac{d(EA)}{dt} = \epsilon_0 A \frac{dE}{dt} \dots \dots \dots (4)$$

Comparing Displacement Current and Real Current:

Comparing Equation $i = \epsilon_0 A \frac{dE}{dt}$ and $i_d = \epsilon_0 A \frac{dE}{dt}$, we see that the real current i charging the capacitor and the fictitious displacement current i_d between the plates have the same value:

$$i_d = i \dots \dots \dots (5)$$

- Thus, the fictitious displacement current i_d can simply be considered as a continuation of the real current i from one plate, across the capacitor gap, to the other plate.
- Because the electric field is uniformly spread over the plates, the same is true of this fictitious displacement current i_d , as suggested by the spread of current arrows in Fig. b.

- Although no charge actually moves across the gap between the plates, the idea of the fictitious current i_d can help us to quickly find the direction and magnitude of an induced magnetic field.

Induced Magnetic Field:

The magnitude of the magnetic field induced by a charging capacitor with parallel circular plates of radius R can be obtained by simply considering the space between the plates to be an imaginary circular wire of radius R carrying the imaginary current i_d . Then, the magnitude of the magnetic field at a point inside the capacitor at radius r from the center is,

$$B = \left(\frac{\mu_0 i_d}{2\pi R^2} \right) r$$

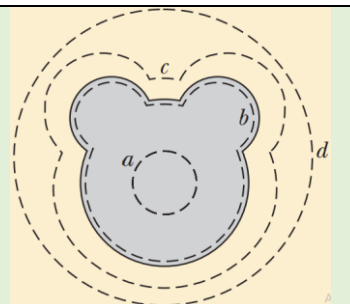
Similarly, the magnitude of the magnetic field at a point outside the capacitor at radius r is

$$B = \frac{\mu_0 i_d}{2\pi r}$$

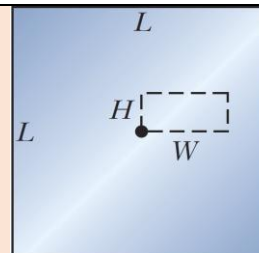
CHECKPOINT: The figure is a view of one plate of a parallel-plate capacitor from within the capacitor. The dashed lines show four integration paths (path b follows the edge of the plate). Rank the paths according to the magnitude of $\oint \vec{B} \cdot d\vec{s}$ along the paths during the discharging of the capacitor, greatest first.

ANSWER: tie of b, c , and d , then a

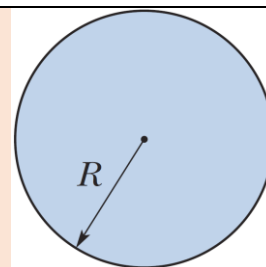
$\oint \vec{B} \cdot d\vec{s} = \mu_0 \epsilon_0 \frac{d\Phi_E}{dt} = \mu_0 i_d$, and thus, depends on the value of i_d .



PROBLEM 32-06: A capacitor with square plates of edge length L is being discharged by a current of 0.75 A . Figure is a head-on view of one of the plates from inside the capacitor. A dashed rectangular path is shown. If $L = 12 \text{ cm}$, $W = 4.0 \text{ cm}$, and $H = 2.0 \text{ cm}$, what is the value of $\oint \vec{B} \cdot d\vec{s}$ around the dashed path?



PROBLEM 32-07: Uniform electric flux. Figure shows a circular region of radius $R = 3.00 \text{ cm}$ in which a uniform electric flux is directed out of the plane of the page. The total electric flux through the region is given by $\Phi_E = (3.00 \text{ mV} \cdot \text{m/s})t$, where t is in seconds. What is the magnitude of the magnetic field that is induced at radial distances (a) 2.00 cm and (b) 5.00 cm ?



PROBLEM 32-13: At what rate must the potential difference between the plates of a parallel-plate capacitor with a $2.0 \mu\text{F}$ capacitance be changed to produce a displacement current of 1.5 A ?

PROBLEM 32-14: A parallel-plate capacitor with circular plates of radius R is being charged. Show that the magnitude of current density of the displacement current is $J_d = \epsilon_0 \left(\frac{dE}{dt} \right)$ for $r \leq R$.

PROBLEM 32-16: A parallel-plate capacitor with circular plates of radius 0.10 m is being discharged. A circular loop of radius 0.20 m is concentric with the capacitor and halfway between the plates. The displacement current through the loop is 2.0 A . At what rate is the electric field between the plates changing?

PROBLEM 32-17: A silver wire has resistivity $\rho = 1.62 \times 10^{-8}\ \Omega \cdot \text{m}$ and a cross-sectional area of 5.00 mm^2 . The current in the wire is uniform and changing at the rate of 2000 A/s when the current is 100 A . (a) What is the magnitude of the (uniform) electric field in the wire when the current in the wire is 100 A ? (b) What is the displacement current in the wire at that time? (c) What is the ratio of the magnitude of the magnetic field due to the displacement current to that due to the current at a distance r from the wire?